Vera Lisovskaja: Licentiate Seminar

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Paper I

This paper concerns combination tests used in adaptive clinical trial designs.

Dragalin is cited:

> *An adaptive design uses accumulating data to decide on how to modify aspects of the study* . . . *without undermining the validity of the trial.*

Stepping back a stage, we should ask why adaptation might occur.

Often, investigators lack important information about objectives or parameter values at the planning stage — and adaptation occurs when this becomes known.

In a group sequential design, early stopping depends on the observed data, as well as the underlying parameters.

Sample size re-estimation methods are often explained in terms of learning about the true treatment effect. I don't think this is why they work (if they do): interim estimates of a treatment effect are very noisy.

Peculiarities of frequent inference

In Section 2.3.1, Vera describes a situation where two clinical trials produce identical data but the two studies lead to different conclusions.

The first trial has a fixed sample size analysis and the data are ***non-significant***.

The second trial uses a combination test: with unequal weighting of observations, this gives a ***significant*** result.

- Group sequential tests exhibit similar properties. Having conducted previous analyses, the requirement for “significance” at a later analysis is higher.
- Studies with multiple hypotheses give more complex examples of the same phenomenon. The scheme for partitioning and re-cycling type I error can have a major effect on which hypotheses are rejected overall.

Bayes methods avoid these issues — but have their own additional requirements. In general, it is good to know that a frequentist method is similar to a Bayes method for some prior.

The dual test

Applying the dual test removes positive outcomes that would lack a certain "credibility".

Examples show that adding this extra hurdle does not greatly decrease power.

Does this mean that there was only a small probability of "non-credible" results to begin with?

NB Conditional on the requirement $Z^N > 1.96$ having an impact, there must be loss of power.

Relation of the dual test to Chen, DeMets & Lan (2004)

There are interesting connections between the dual test and the proposal of Chen, DeMets & Lan (hereafter CDL).

We can regard the original fixed sample test in CDL's set-up as a combination test with the usual weights, based on sample sizes.

An estimate $\widehat{\theta}$ of the treatment effect θ is observed at an interim analysis.

In cases where conditional power under $\theta = \widehat{\theta}$ is greater than 0.5, CDL suggest increasing sample size and using the "standard" significance test at the end. Thus, for a 2.5% one-sided test, H_0: $\theta \leq 0$ is rejected if $Z^N > 1.96$.

CDL show these adaptations can only decrease the type I error rate. In effect, they are using the dual test in these cases.

They refer to cases where conditional power under $\theta = \widehat{\theta}$ is greater than 0.5 as the "promising zone" and they restrict increases in sample size to such cases.

Relation of the dual test to Chen, DeMets & Lan (2004)

Thus, CDL favour adaptation where the final decision follows a “standard” hypothesis test, rejecting H_0 if $Z^N > 1.96$.

For a method to be only a little conservative, and so not lose much power, Vera wants decisions to be based on the combination test part of the dual test, $Z^W > 1.96$.

- The requirement $Z^N > 1.96$ is also present in the dual test but it does not have an additional effect in these cases.
- In their construction, CDL do not permit such cases to happen!

CDL suggest it is good to increase sample size in the “promising zone” as this is where additional power can be gained (but they don’t really justify this claim).

When ***is it*** best to take additional observations?

Choice of sample-size rule (SSR)

In Section 2.4, Vera discusses various sample-size rules.

Most of these rules are designed to respond to interim information about the unknown treatment effect θ — but there is not much information!

Consider a test of H_0: $\theta \leq 0$ vs $\theta > 0$ with one-sided type I error 0.025 and power 0.9 at $\theta = \delta$.

Calculations show that at an interim analysis with half the final data, a 95% confidence interval for θ is

$$\widehat{\theta} \pm 0.856\,\delta.$$

It is naive to conclude on observing $\widehat{\theta} = 0.5\delta$ that the true θ must be half the target effect size, δ — the interim data are consistent with true effects $\theta > \delta$, and with effect sizes $\theta < 0$!

Choice of sample-size rule: Decision theory

A general decision theory approach allows an informed choice of second stage sample size

- Balancing increased power against additional sample size,
- Allowing for uncertainty in the interim estimate $\widehat{\theta}$,
- Knowing the current data that will contribute to the final test.

Mehta & Patel (2006) use decision theory with net present value as the gain function, and so include a penalty for time taken to reach a positive outcome.

We can use the same general framework for other objectives, e.g., a gain function of

$$Pr\{\text{Reject } H_0\} - E(\text{Sample size}).$$

Optimising the SSR for a Bayes decision problem can provide an optimal design in frequentist terms.

Choice of sample-size rule (SSR)

Mehta & Pocock (*Statistics in Medicine*, 2011) use the CDL construction to create designs that increase sample size in the "promising zone".

They describe an example of a clinical trial comparing treatments for schizophrenia.

- Response is measured 26 weeks after the start of treatment.
- Thus, at an interim analysis there are patients who have been treated but whose responses will only be observed later.
- This means that traditional group sequential designs are not necessarily appropriate.

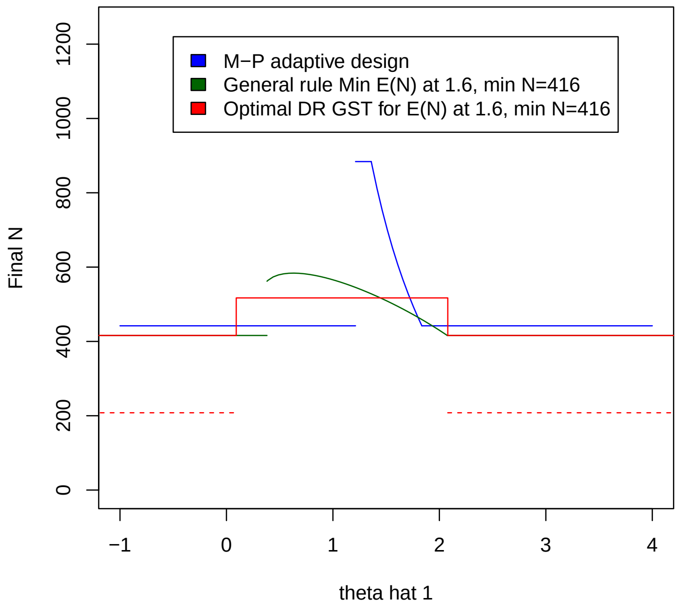
Bruce Turnbull and I have assessed designs for this problem.

We have found optimal SSRs for a specified type I error probability and power curve.

Sample size rules

The figure shows sample size rules for

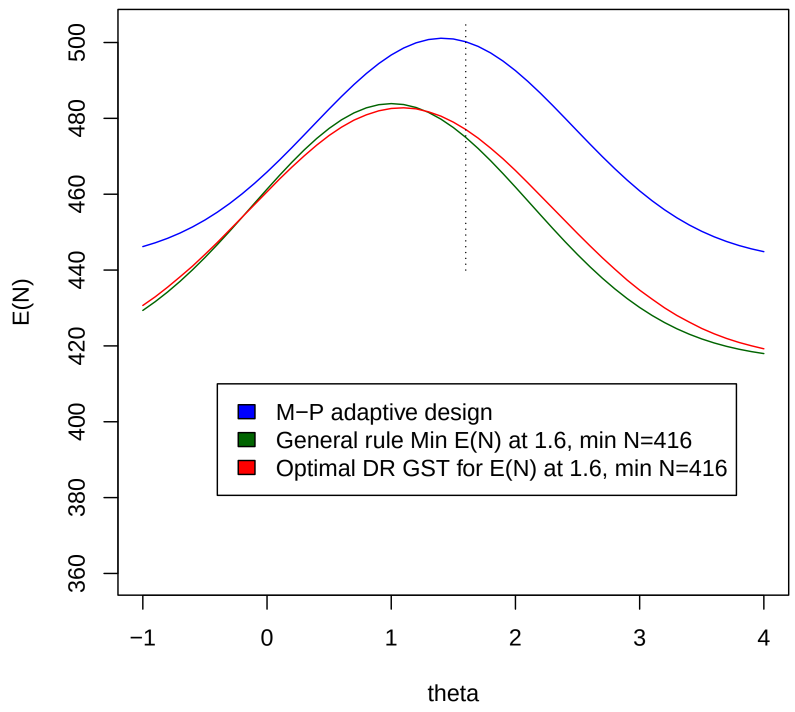
- Mehta & Pocock's implementation of the CDL method.
- The SSR minimising $E_{\theta=1.6}(N)$, dealing appropriately with the "pipeline" subjects at the interim analysis,
- A "Delayed Response Group Sequential Test" (Hampson & Jennison, 2012).



Expected sample size curves for three SSRs

The curves in the figure show it is possible to improve on the design proposed by Mehta & Pocock.

Similar curves arise for the optimal adaptive SSR and the “Delayed Response Group Sequential Test” which has a fixed second group size.



Returning to the dual test

1. Hampson & Jennison (2012) found their optimisation procedure gave designs with one-sided type I error rate 0.025 that had a final critical value for Z below 1.96.

They imposed a condition, as in the dual test, to avoid this.

2. Are there other situations where the dual test principle can be usefully applied?

- Adaptive designs with multiple hypotheses, using a closed testing procedure and combination tests across stages.
- Inference on termination after an adaptive trial.

3. If we know the dual test requirement is to be imposed, it should be built into the process of choosing an efficient trial design. Can this make a noticeable difference?

Paper II

This paper concerns choice of the dose to test in a Phase III clinical trial — and when it may be beneficial to test two doses.

There are many ingredients, starting with:

- Gain function: Success if at least one dose is shown to be effective and safe in one Phase III trial,
- E_d: The event that dose d is shown to be effective,
- S_d: The event that dose d is shown to be safe,
- A model for $Pr\{E_d\}$,
- A model for $q(d) = Pr\{S_d \mid E_d\}$,
- Assumption of independence of events E_d and S_d *(conditional on certain model parameters)*.

I have difficulty with the assumptions about E_d and S_d . . . .

Comments on assumptions

Recall the definitions

- E_d: The event that dose d is shown to be effective,
- S_d: The event that dose d is shown to be safe.

Consider

$$q(d) = Pr\{S_d \,|\, E_d\}. \tag{1}$$

It is reasonable for $Pr\{E_d\}$ to increase with d and $Pr\{S_d\}$ to decrease with d.

We expect these rates of increase and decrease to be related, hence the interest in $q(d)$ defined in (1). Perhaps $q(d)$ is considered a quantity that can be "elicited"?

Questions

What is the probability of the event S_d in (1) meant to capture?

Is it the randomness in whether the dose is actually safe?

Or is it uncertainty as to whether safety problems are detected in Phase III?

Comments on assumptions

Suppose we take the definition

$$q(d) \;=\; Pr\{S_d \,|\, E_d\}.$$

Note that the conditioning is on the true efficacy at dose d (if this is treated as random) and on *observed responses at dose* d.

This implies that $q(d)$ depends on the numbers of subjects observed at dose d and on placebo.

This does not strike me as a natural choice of quantity to model.

Moreover, when sample size is varied later, it appears that $q(d)$ remains the same.

Another approach to modelling

Efficacy

Take the example of the E_{max} model with mean response function

$$\mu(d) \;=\; E_0 + E_m \frac{d^\gamma}{d^\gamma + ED_{50}^\gamma},$$

and responses at dose d

$$Y_{di} \;\sim\; N(\mu(d),\, \sigma^2), \quad i = 1,\, 2,\, \ldots\, .$$

Safety

Specify a model for, say, the incidence rate of serious adverse events

$$\rho(d) \;=\; \rho_{max}\, \Phi\{a(d-b)\}.$$

Joint modelling

Link parameters $(E_0, E_m, ED_{50}, \gamma)$ and (ρ_{max}, a, b) either by deterministic relationships or through a joint prior distribution.

Results in Paper II

The results in the paper are fascinating.

Two doses in Phase III

The two dose option is shown to be beneficial only when there is uncertainty about a model parameter — which is typically the case.

Then, the best choice depends on the distribution of the unknown parameter.

How sensitive is this “optimal” choice of doses to the specified distribution for the unknown parameter? Do things go badly wrong if this distribution is mis-specified?

Also, with two active doses, would it be better to allocate more subjects to placebo?

Choice of multiple testing procedure

Model assumptions about (various forms of) monotonicity play an important role.

Can we make this design group sequential? (Ultimately, we may blur the distinction between Phases II and III.)

Conclusion

The two papers offer a contrast.

Paper I gives a very clean and tidy account of a research area that is close to complete (if not there already).

This is nice, but there may not be so much to discuss!

Paper II reports work on a challenging problem which raises new questions as it answers others.

It is fine to make assumptions and see where they lead: indeed, it is important to learn whether a problem with simple assumptions is actually solvable.

Then, one can re-visit the problem formulation with a better idea of just how complex we might allow it to become.